Optimization — Homework 3

Solutions

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Notation. Vector inequalities are componentwise. A_j denotes column j of A, and a_i^{T} denotes row i.

Problem 1. Computing the dual

The primal is

$$\begin{aligned} \min \quad & x_1 - x_2 \\ \text{s.t.} \quad & 2x_1 + 3x_2 - x_3 + x_4 \le 0, \\ & 3x_1 + x_2 + 4x_3 - 2x_4 \ge 3, \\ & -x_1 - x_2 + 2x_3 + x_4 = 6, \\ & x_1 \le 0, \quad x_2, x_3 \ge 0, \quad x_4 \text{ free.} \end{aligned}$$

Let A be the matrix of the three general constraints, $b = (0, 3, 6)^{\mathsf{T}}$, and $c = (1, -1, 0, 0)^{\mathsf{T}}$. Using the Lagrangian

$$L(x, p) = c^{\mathsf{T}}x + p^{\mathsf{T}}(b - Ax),$$

choose $p_1 \le 0$, $p_2 \ge 0$, and p_3 free, so that $L(x, p) \le c^{\mathsf{T}}x$ for every primal-feasible x.

The dual function is

$$g(p) = p^{\mathsf{T}}b + \inf_{x_1 \le 0,\, x_2, x_3 \ge 0,\, x_4 \in \mathbb{R}} \sum_{j=1}^{4} (c_j - p^{\mathsf{T}}A_j)x_j.$$

This infimum is finite precisely when

$$c_1 - p^{\mathsf{T}}A_1 \le 0, \qquad c_2 - p^{\mathsf{T}}A_2 \ge 0, \qquad c_3 - p^{\mathsf{T}}A_3 \ge 0, \qquad c_4 - p^{\mathsf{T}}A_4 = 0.$$

Under these conditions the infimum is 0, attained at $x = 0$; otherwise it is $-\infty$. Thus maximising $g(p)$ gives

Answer.

$$\begin{aligned} \max \quad & 3p_2 + 6p_3 \\ \text{s.t.} \quad & 2p_1 + 3p_2 - p_3 \ge 1, \\ & 3p_1 + p_2 - p_3 \le -1, \\ & -p_1 + 4p_2 + 2p_3 \le 0, \\ & p_1 - 2p_2 + p_3 = 0, \\ & p_1 \le 0, \quad p_2 \ge 0, \quad p_3 \text{ free.} \end{aligned}$$

Problem 2. A theorem of the alternative

For $A \in \mathbb{R}^{m\times n}$, prove the equivalence of:

(i) $Ax \geq 0$ and $x \geq 0$ imply $x_1 = 0$;

(ii) there exists $p \geq 0$ such that $p^{\mathsf{T}}A \leq 0$ and $p^{\mathsf{T}}A_1 < 0$.

Proof of (ii) $\Rightarrow$ (i). Let p satisfy (ii), and suppose $Ax \geq 0$ and $x \geq 0$. Then

$$0 \leq p^{\mathsf{T}}Ax = \sum_{j=1}^{n}(p^{\mathsf{T}}A_j)x_j \leq (p^{\mathsf{T}}A_1)x_1 \leq 0.$$

The first inequality uses $p \geq 0$ and $Ax \geq 0$; the remaining inequalities use $p^{\mathsf{T}}A_j \leq 0$ and $x_j \geq 0$. Consequently $(p^{\mathsf{T}}A_1)x_1 = 0$. Since $p^{\mathsf{T}}A_1 < 0$, we obtain $x_1 = 0$. □

Proof of (i) $\Rightarrow$ (ii). Consider the primal–dual pair

$$\text{(P)} \quad \min_{x\geq 0} \; -x_1 \quad \text{s.t. } Ax \geq 0,$$

$$\text{(D)} \quad \max_{p\geq 0} \; 0 \qquad \text{s.t. } A^{\mathsf{T}}p \leq -e_1,$$

where $e_1 = (1, 0, \ldots, 0)^{\mathsf{T}} \in \mathbb{R}^n$. The dual constraints follow from the primal objective coefficient $-e_1$, the nonnegative variables x, and the zero right-hand side.

The primal is feasible at $x = 0$. By (i), every feasible x has $x_1 = 0$, so its optimal value is 0 and is attained. Strong LP duality therefore guarantees a feasible dual solution p. It satisfies

$$p \geq 0, \qquad p^{\mathsf{T}}A_1 \leq -1 < 0, \qquad p^{\mathsf{T}}A_j \leq 0 \quad (j = 2, \ldots, n),$$

which proves (ii). □

Problem 3. A small infeasible subsystem

Let $A \in \mathbb{R}^{500\times 100}$ and suppose $Ax \le b$ is infeasible. We prove that at most 101 of its inequalities already form an infeasible system.

1. Introduce a common violation variable. Consider

$$\text{(P)} \qquad \min_{x\in\mathbb{R}^{100},\, t\in\mathbb{R}} \; t \quad \text{s.t.} \quad a_i^{\mathsf{T}} x - t \le b_i, \quad i = 1, \dots, 500.$$

This LP is feasible: set $x = 0$ and take t sufficiently large. Every feasible pair has $t > 0$, since $t \le 0$ would imply $Ax \le b$. Thus the objective is bounded below. A feasible LP with a finite lower bound attains its optimum, so there is an optimal pair $(x^\star, t^\star)$ with $t^\star > 0$.

2. Choose an optimal dual solution with small support. Writing the constraints as $-a_i^{\mathsf{T}} x + t \ge -b_i$ gives the dual

$$\text{(D)} \qquad \max_{p} \; -b^{\mathsf{T}} p \quad \text{s.t.} \quad A^{\mathsf{T}} p = 0, \quad \mathbf{1}^{\mathsf{T}} p = 1, \quad p \ge 0,$$

where $\mathbf{1} \in \mathbb{R}^{500}$ is the all-ones vector. Strong duality gives an attained dual optimum equal to $t^\star$.

The dual feasible set is a nonempty compact polytope, since it is a closed subset of the simplex $\{p \ge 0 : \mathbf{1}^{\mathsf{T}} p = 1\}$. A linear objective on this polytope has an optimal extreme point $p^\star$. At an extreme point of a standard-form polyhedron, the equality-matrix columns corresponding to positive variables are linearly independent. Here the equality matrix is

$$\begin{pmatrix} A^{\mathsf{T}} \\ \mathbf{1}^{\mathsf{T}} \end{pmatrix} \in \mathbb{R}^{101\times 500}.$$

Therefore the support $S = \{i : p_i^\star > 0\}$ satisfies $|S| \le 101$.

3. Prove that the inequalities in S are infeasible. Complementary slackness for $(x^\star, t^\star)$ and $p^\star$ gives

$$p_i^\star (b_i - a_i^{\mathsf{T}} x^\star + t^\star) = 0.$$

Hence $b_i = a_i^{\mathsf{T}} x^\star - t^\star$ for $i \in S$, and the dual equalities yield

$$\sum_{i\in S} p_i^\star b_i = \left(\sum_{i\in S} p_i^\star a_i\right)^{\mathsf{T}} x^\star - t^\star \sum_{i\in S} p_i^\star = -t^\star < 0.$$

If some x satisfied $a_i^{\mathsf{T}} x \le b_i$ for every $i \in S$, then

$$0 = \left(\sum_{i\in S} p_i^\star a_i\right)^{\mathsf{T}} x \le \sum_{i\in S} p_i^\star b_i = -t^\star < 0,$$

a contradiction. Thus the subsystem indexed by S is infeasible and contains at most 101 inequalities. The remaining inequalities can be discarded while preserving infeasibility. Figure 1 illustrates the result in two dimensions. □

Illustration and code for Problem 3

For $n = 2$, the three inequalities $x_2 \le 0$, $x_2 \ge x_1 + 1$, and $x_2 \ge -x_1 + 1$ are jointly infeasible, although every pair is feasible. Thus the bound $n + 1 = 3$ is attained in this example.

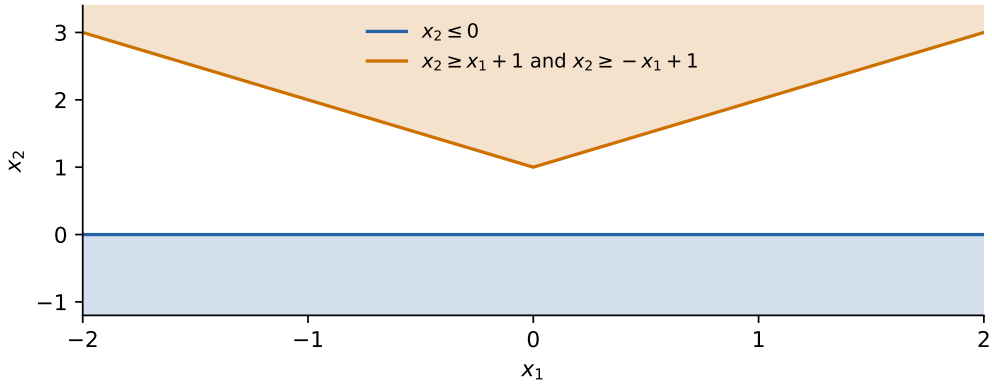


Figure 1: The blue half-plane and the orange intersection of the other two half-planes are disjoint. With rows ordered as above, $p = (2, 1, 1)^\mathsf{T}$ satisfies $p \ge 0$, $A^\mathsf{T} p = 0$, and $b^\mathsf{T} p = -2 < 0$, certifying infeasibility.

The following Python code generates the figure and checks the certificate and pairwise feasibility using integer arithmetic. It illustrates the general proof above.

```
from pathlib import Path
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt

A = np.array([[0, 1], [1, -1], [-1, -1]])
b = np.array([0, -1, -1])
p = np.array([2, 1, 1])
assert (p >= 0).all() and (A.T @ p == 0).all() and b @ p < 0
# Witnesses after omitting rows 0, 1, and 2, respectively.
for omitted, x in enumerate([[0, 1], [1, 0], [-1, 0]]):
    keep = np.arange(3) != omitted
    assert (A[keep] @ x <= b[keep]).all()
print("Certificate: A.T @ p =", A.T @ p, "; b @ p =", b @ p)
print("Every pair of inequalities is feasible: True")

u = np.linspace(-2, 2, 401)
fig, ax = plt.subplots(figsize=(6.4, 2.5), layout="constrained")
ax.fill_between(u, -1.2, 0, color="#2463A6", alpha=0.2)
ax.fill_between(u, 1 + abs(u), 3.4, color="#CE7000", alpha=0.2)
ax.axhline(0, color="#2463A6", label=r"$x_2\leq 0$")
ax.plot(u, 1 + abs(u), color="#CE7000",
        label=r"$x_2\geq x_1+1$ and $x_2\geq -x_1+1$")
ax.set(xlim=(-2, 2), ylim=(-1.2, 3.4), xlabel=r"$x_1$",
       ylabel=r"$x_2$", xticks=[-2, -1, 0, 1, 2])
ax.legend(loc="upper center", fontsize=9, frameon=False)
ax.spines[["top", "right"]].set_visible(False)
out = Path(__file__).resolve().parent / "figures"
out.mkdir(exist_ok=True)
fig.savefig(out / "p3_focused_example.pdf")
plt.close(fig)
```